MILITARY SPECIFICATION

KETTLES, HEATING, BITUMEN, TRAILER-MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two sizes of diesel-fuel trailer-mounted kettles.

1.2 Classification. The kettles shall be of the following sizes, as specified:

Size 1	- 80-gallon capacity.
Size 2	- 165-gallon capacity.

2. APPLICABLE DOCUMENTS

2*1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-E-915	- Extinguisher, Fire, Dry Chemical (Hand Portable).
VV-F-800	- Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Asociated Equipment).
MIL-P-514	- plates, Identification, Instruction and Marking, Blank.
MIL-D-771	- Distributors, Bitumen and Water, Packaging of.
MIL-T-704	Treatment and Painting of Materiels.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-13486/12	- Cable-, Special Purpose, Electrical, Low Tension, Heavy Duty Multiple Conductors Unshielded.
MIL-I-22023	- Insulation Felt, Thermal and Sound Absorbing Felt, Fibrous Glass, Flexible.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-E-52929	- Enamel, Alkyd, Camouflage, Flash Dry.

STANDARDS

MILITARY

MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-209	Sling and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-889	Dissimilar Metals.
MS35844	Plug, Machine Thread, Magnetic: Drain.
MS75020	Connector, Plug, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF' TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the U.S. Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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(Copies of specifications and standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualification.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D5 - Method of Test for Penetration of Bituminous Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC. (TTMA)

TTMA Recommended Practices

TTMA Technical Bulletins

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., Suite 203, 2430 'Pennsylvania Ave., Washington, DC 20037.)

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to Tire and Rim Association, Inc., Command Building, 34 North Hawkins Avenue, Akron, Ohio 44313.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The kettles shall consist essentially of a gasoline engine, bitumen pump, melting tank with an outside furnace shell, removable-torch-type fuel burner with fuel tank, and hand-spray-assembly mounted on a two-wheel pneumatic-tired cart type trailer. All controls, accessories, and other components necessary for all operations specified herein shall be provided. Except as specified herein, the kettle automotive type components, assemblies and accessories shall be commercial automotive products and they shall comply with all Federal Motor Vehicle Safety Standards as specified herein, applicable on the date of manufacture to nonmilitary, nonexport motor vehicle products.

3.1.1 Dimensions. The kettle shall conform to the dimensional requirements specified in table 1.

TABLE 1. Dimensions.

Overall length	130 inches, maximum
Overall width	70 inches, maximum
Overall height (shipping)	70 inches, maximum
Ground clearance (exclusive of suspension)	12 inches, minimum
Towbar eye height range (see J849)	31 thru 37 inches
Shipping weight (maximum)	2200 pounds

3.2 First article (reproduction model). Unless otherwise specified (see 6.2) the contractor shall furnish one or more kettles for examination and testing within the time frame specified, to prove prior to starting production that his production methods and choice of design detail will produce kettles that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.5). When specified (see 6.2) the Government will conduct any or all of the preproduction examination and tests.

3.3 Material. Material shall be as specified herein (see 6.8). Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials and special treatments utilized shall provide resistance to corrosion. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other unless the metals are protected against galvanic corrosion.

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@ 3.3.1 Material deterioration prevention and control (MADPAC). The kettle shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of material deterioration. Assurance shall be given in the form of a finish plan which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of material deterioration in the applicable operating and storage environments to which they may be exposed.

3.4 Safety and human factors.

3.4.1 Safety. All moving parts which are hazardous to operator or maintenance personnel shall be fully enclosed or guarded. All parts which are subject to high operating temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated to the extent necessary to eliminate the hazard. Fuel tanks shall be located in a manner which will not allow spills and overflows to run onto enigine, exhaust, furnace shell, or electrical equipment. Electrical equipment shall be insulated, fully enclosed, or guarded. Nonfunctional sharp edges, projecting points, and protruding length more than two full threads of fastening devices shall be avoided. Exhaust from the kettle furnace. and engine shall be directed so that they do not endanger personnel or obstruct the view of the operator.

I 3.4.2 Fire extinguisher. When specified, (see 6.2) a dry chemical 2-1/2 pound fire extinguisher with a U. L. rating of 8-B:C conforming to O-E-915 and mounting bracket will be furnished by the Government (see 6.4). The contractor) shall mount the extinguisher on the kettle in an area that is accessible from the operators station.

3.4.3 Human factors. The characteristic of the kettle shall provide for operation by personnel ranging from the small man clothed, through the large man clothed in accordance with SAE J833.

3.4.4 Noise level. The noise level of the kettle shall be 85dBA or less at the operator's station when measured in accordance with SAE J919.

3.5 Maintenance.

3.5.1 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed, Covers or plates which are normally removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be located for accessibility. The configuration of the kettle shall provide maximum accessibility for removal of engine and drive train components. All items of material intended to be carried on board the kettle as on equipment material shall be conveniently located, adequately mounted to prevent damage during transport and mounted in such a position that they will

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not block access to any assembly, component or part during maintenance operations. Ease of maintenance provisions shall incorporate features insuring operating clearances for hand tools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase, and adding oil to all gear cases.

3.5.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Sizes and types of common hardware and fastening devices, such as bolts, nuts, screws, washer, connectors, and valves shall be held to a minimum.

3.6 Trailer: The kettle shall be mounted on a pneumatic-tired, 2-wheel, cart-type trailer chassis that shall provide the on-highway and secondary road mobility specified as follows:

3.6.1 Mobility. The kettle shall perform as specified when: (1) equipped with all attachments, equipment, and accessories stowed for travel and with all tanks filled to capacity; (2) towed over highways, and secondary roads by a truck matched to the kettle.

3.6.1.1 Highway travel. The kettle shall travel over smooth, hard surfaced, primary highways at sustained speeds up to 45 miles per hour (mph) without failure or permanent deformation.

3.6.1.2 Secondary road travel. The kettle shall travel over gravel and improved roads at speeds up to 25 miles per hour (mph)without failure or permanent deformation.

3.6.2 Trailing and turning ability. The kettle shall trail the towing vehicle without bottoming, cramping, weaving, or sidesway. The kettle shall follow the truck without exceeding the tracking deviation, as specified in DOT Federal Motor Carrier Safety Regulations, 393.70, when towed over a paved surface. Jackknife turns of at least 60 degrees, as measured from straight ahead to the right and to the left, shall not cause damage to or interference between the truck and kettle.

3.6.3 Frame. The frame shall be of the ladder type with longitudinal frame side rails reinforced with transverse crossmembers and gussets to meet the transport and the equipment operating conditions specified herein. The web of the side rails shall be continuous from the towbar to the last crossmember. Gusseted crossmembers shall tie together, maintain alinement, and distribute the stress of the suspension mounts.

3.6.4 Suspension. The kettle shall be equipped with a sprung, single axle suspension system, tailored to the kettle. The suspension manufacturer's rated capacity shall be not less than 150 percent of the load imposed on each member with the kettle loaded to maximum gvw. When tested as specified in 4.5.2.10, the suspension shall provide a nondamaging ride on graveled surface secondary

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roads and a stable ride on paved highways. The installation of the suspension shall be in correct alinement with the frame and with the axle so that the two diagonal measurements from the center of the lunette to these suspension components do not differ in excess of l/16-inch tolerance for every 10 feet or fraction thereof. Elastomer cushioned bumper stops or blocks shall be furnished to prevent axle-to-frame contact and limit suspension articulation at full compression (jounce) and rebound.

3.6.5 Axle assembly. The axle shall be rated by the axle manufacturer at a continuous load capacity rating of not less than 150 percent of the kettle weight when loaded to maximum gvw. The axle hubs shall be of the 5-stud, 4.50-inch bolt circle type.

3.6.6 Wheels, rims, and tires. The kettle shall be equipped with single remountable disc wheel and tire assemblies. The wheel shall be size 16.00 - 4.50 EO, safety rim drop center type having 1.9 inch offset. The disc wheels shall be of the heavy duty type with T&RA rim. The rim and wheel disc shall be constructed by the same wheel manufacturer and shall be in accordance with the wheel manufacturer's recommendations for the loads and service specified herein. Rims, tires, inner tubes, valve cores, and caps shall be in accordance with the T&RA Yearbook and Supplementary Service Data Book. The tires shall be of not less than 100 level quality and shall be the tube or tubeless type, having a minimum size of 7.00 by 16, load range C (6 ply rating). The load imposed on each tire, measured on level ground under loaded travel conditions, shall not exceed the recommended rated loads and inflation pressures of the tires for 55 mph highway speed for continuous highway operation. Valve caps shall be of metal and shall be equipped with elastomer inner seals. Valves shall be of the C-4, heat resistant types.

3.6.6.1 Balancing. Tires, wheels, and hubs shall be inherently in-balance, singly and collectively after assembly, to preclude wheel tramp, drumming, or excessive tire wear, when operated up to the maximum highway speeds specified here in.

3.6.7 Fenders. The kettle shall be equipped with two fenders. The fenders shall restrict water and other road surface substances thrown from the kettle wheels. The fenders shall be either contoured to the normal axle centers or shall be octagonal shape about the wheels and shall include inner splash shields. The inner splash shield shall protect the tires from the heat radiated from the kettle furnace. The fenders shall support a weight of not less than 200 pounds imposed on any l-square-foot area at any time without damage or permanent deformation. The outer edges of the fenders shall be rolled to reduce side road splash.

3.6.8 Rear bumper. The kettle shall be equipped with a horizontal rear bumper extending the full width of the kettle. The bumper shall extend rearward not less than 4 inches from any part of the kettle. The left and right end of the bumper shall be rounded and closed to prevent hooking when sideswiped by an obstruction.

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3.6.9 Intervehicular connection. When a vehicular electrical system is specified (see 6.2) the kettle shall be furnished with an intervehicular electrical harness as specified (see 6.2) for connecting the towing vehicles, electrical system to the kettle's systems. The harness shall be installed on the kettle so that it will extend forward between 36 and 42 inches toward the towing vehicle, measured from the harness plug to the towbar with the harness doubled back on the kettle. The harness shall be routed or protected so that dropping of the tailgate of the towing vehicle shall not cause physical damage. When a 12 and 24 volt vehicular electrical system is specified, intervehicular harness compatible with each system shall be provided as specified herein.

3.6.10 Towbar components. The towbar shall be constructed to embody structural strength to meet or exceed the following requirements: support and transport the weight of the kettle at maximum gvw; transmit and distribute the imposed load to the suspension system and eye, and resist bending and twisting without permanent deformation or part failure when subjected to the performance requirements specified herein. The towbar eye shall conform to SAE J847 and shall have l/8-inch tolerance of the longitudinal centerline of the kettle. The towbar shall conform to the height range specified in SAE J849 (31 through 37 inches). The lunette (towbar eye) shall be inserted in a support that is welded into the towbar. The lunette support shall prevent lunette rotations. Two safety chains, equipped with safety type grabhooks, shall be furnished and shall conform to SAE J697 and to DOT Federal Motor Carrier Safety Regulations as specified herein. The forward length of each chain hook shall extend forward 36 to 42 inches toward the towing vehicle, measured from the center of the towbar eye. The safety chain hook shall pass through a shackle with a 2-3/4 inch diameter opening. The attachment end of the safety chains shall be either perimeter welded or bolted to the towbar.

3.6.11 Landing gear. A single-leg landing gear with capacity to fulfill the requirements specified herein, shall be installed on the towbar of the kettle. The landing gear shall be vertically adjustable for not less than 12 inches. The landing gear shall either retract or be hinged and fold to provide not less than the ground clearance of the first trailer crossmember. The landing gear shall be arranged for positive locking into park and travel positions. A folding-type crank handle for operating the vertical adjustment mechanism shall be furnished. A travel retainer for the handle shall be provided. The tangential driver's effort at the handgrip to raise the maximum gvw load shall not exceed 30 pounds. The mechanism shall be protected from exposure to foreign matter which would impair functional efficiency. The landing gear shall incorporate a self-alining pad which will provide a ground bearing load not exceeding 20 pounds per square inch when the maximum towbar weight is imposed. The edges of the pad shall be turned up.

3.6.12 Rear support leg. An adjustable support leg shall be furnished and shall be located at the left or right rear corner of the kettle. The support leg shall be in accordance with the minimum requirements as specified for the landing gear in 3.6.11.

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3.6.13 Chock blocks. A chock block conforming to figure 1 and a cradle conforming to figure 2 shall be provided for each wheel. The captive chains shall permit placing the chock block both to the front and to the rear of each wheel.

3.6.14 Vehicular electrical system. When specified (see 6.2) the kettle shall be equipped with a waterproof, 12-volt vehicular electrical system. Except as specified herein, the electrical system shall be in accordance with the following requirements:

a. DOT Federal Motor Vehicle Safety Standards (FMV SS).
b. DOT Federal Motor Carrier Safety Regulations (FMCSR).
c. SAE Handbook, Electrical Equipment and Lighting Section.
d. TTMA Recommended Practices RP No. 6, RP No. 9, RP No. 14, RP No. 40, and Technical Bulletin No. 65.

All components shall be provided with the maximum practical protection from damage, shock, and vibration through means of location and resilient mounts. The electrical components shall be in accordance with table II. Item quantities that are not specified shall be furnished as required (AR) for each drier-mixer.

TABLE II. Vehicular electrical system.

Description	Requirements	Quantity Per Heater
7 conductor plus and intervehicular harness assembly	SAE J560 and TTMA RP#40, 7-wire type with plated metal connectors, splash proof plug cover. SAE J1067 seven conductor jacketed cable fastened to connection box.	1 ea
Connection box	Splash proof type with stud and nut terminals.	1 ea min
Circuit breakers	SAE J553 protecting all live circuits.	AR
Wiring	Heavy duty type. Wire size shall conform to SAE J555.	AR
Wire identification	SAE 3560 or J821.	AR
Wiring harness	Polyethylene electrical conduit, each end weather proofed with silicone sealant or with one-half over-lapping turns of plastic electrical tape.	AR

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TABLE II. Vehicular electrical system. (continued)

Description	Requirements	Quantity Per Heater
Grommets	Provided where needed along the wire routing to prevent chafing.	AR
Supports	Elastomer cushioned support straps or elastomer ties with mounts; wiring support spacing 36 in. max. span, prevent whipping, chafing, and fatigue damage.	AR
Wiring splices	Mechanical splicing devices; interior waterproofed with silicone lube or compounds.	AR
Connectors and terminal	Waterproof types; blade type SAE J858, pin type SAE J928, and spade SAE J561. Blade and pin types fully enclosed and lubricated, prior to assembly with silicone lubricant; spade type equipped with heat shrinkable sleeves and coated witin electrical varnish after assembly; all connectors and terminals plated or solder-coated to retard corrosion.	AR
Connector holder	Plastic or plastic-insulated metal spring clips; fastened to vehicle; furnished to anchor wire-to-wire connectors.	1 ea per connector
Running lamps	4 sidemarker, 4 clearance, 3 rear identification; per TTMA RP-108/l or RP-108/3. Waterproof and shock resistant types.	11 ea min
Combination lamps	2 turn and tail outboard and 2 stop and tail inboard; per TTMA RP No. 14 and SAE J759; double contact; 3 wire type including ground wire; waterproof and shock resistant type.	4 ea min

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TABLE II. Vehicular electrical system. (continued)

Description	Requirements	Quantity Per Hea e
Lamp lenses and bulbs	Heavy duty or fleet service type bulbs, bulbs mounted in vibration-protected sockets with insulators end stainless steel or rubber bulb tension devices.	AR
Reflectors	2 on each corner, round type, securely fastened by bolts, screws or rivets.	8 ea min
Wiring diagram	Plate furnished on trailer per TTMA Bulletin #65.	1 ea

3.6.14.1 Conversion to 24 volt electrical system. When specified (see 6.2), the kettle trailer shall be equipped with a military compatible 24-volt service lighting system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the TTMA 7-contact plug, the trailer shall be equipped with the following military intervehicle components:

a. Military 12-contact plug - MS75020-l.
b. Military vehiclular cable - MIL-C-13486/12 or SAE JI067 seven conductor jacketed cable.

The military plug and cable or SAE cable shall be the same length as the TTMA plug and cable and both shall be labeled with identification plates for 24-volts and 12-volts respectively, negative ground. The military 24-volt inter-vehicular cable shall be routed into a splashproof stud and nut type connection box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. All unused leads of the cable shall be insulated and sealed watertight. The military 24-volt circuits shall be connected through replaceable resistors, for live circuits, and to the trailer frame, for ground circuits, in accordance with table III.

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TABLE 111. Military-commercial connections.

Function	Mil. Plug Terminal	Connection Instructions	Comm. TTMA Wiring Ident.
Blackout marker	A	No.	-
Ser. left turn & stop	B	Connect to -	YEL (3)
Blackout marker	C	No.	-
Ground	D	Connect to -	WHT (1)
Ser. tail & marker	E	Connect to -	BLK (2)
		& Connect to -	BRN (6)
Blackout stop	F	No.	-
Blackout marker	R	No.	-
Ser. right turn & stop	J	Connect to -	GRN (5)
Aux. power (hot)	K	No.	BLU (7)
Ground	L	Ground to Frame	-
Electric brake	M	No.	-
Electric brake	N	No.	-

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically isolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits letter coded B and J on the tactical trucks are passenger car combination stop and turn signal type, using the same lamp filament, and there is no single stop lamp circuit. As a result, the trailer's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the trailer's normal stop circuit, RED (4), cannot be energized from the military 24-volt system input.

3.7 Heating system. The kettle shall be equipped with a portable generating type heating system as specified herein.

3.7.1 Burner. The burner shall be removable-generating-torch type and shall burn without smoking, not less than 2.5 gallons per hour of diesel fuel conforming to VV-F-800. The burner shall be equipped with a regulating valve, fuel strainer, windshield or baffle, fuel line, and hand grips so burner can be used as a portable type torch. The fuel line shall be an oil- and heat-resisting, flexible, synthetic rubber hose, and shall be equipped with ground-joint fittings. The hose shall be long enough to allow moving the burner to beat the asphalt pipes and pump. The burner mounting bracket shall support the burner and shall hold the burner during road travel as specified herein. A fuel line rack shall be provided that will store and protect the fuel line when in the transporting position.

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3.7.2 Burner fuel tank. The burner fuel tank shall have a capacity of not less than 20 gallons, shall be made of steel, and shall be constructed in accordance with the Boiler and Pressure Vessel Code for a working pressure of 100 pounds per square inch (psi). Presence of the ASME official code U-symbol stamped or marked on the fuel tank and the furnishing of a copy of the ASME data sheet (Form U-1) may be accepted as evidence that the fuel tank conforms to the ASME Rules for Construction of Unfired Pressure Vessels. The tank shall be equipped with a 2-inch filler opening having a threaded cap; manually-operated, air-release valve; an air pressure gage calibrated in 5-pound increments from 0 to 60 pounds; a hand-operated air pump; and a safety valve set for not more than 55 psi. The cylinder of the air pump shall be located inside the fuel tank. The outlet to the fuel burner shall be equipped with a globe valve. The valve shall have threaded connections. The tank shall be mounted on the chassis of the trailer and shall be protected from the heat radiated by the melting tank and the burner. A caution plate shall be located near the fuel tank filler opening. The caution plate shall read: "FILL WITH DIESEL FUEL ONLY". The plate shall be anodized aluminum, sunken-etched or raised-relief. The plate shall be distinguished by the words "CAUTION" or "WARNING" in red lettering (see 3.12.2). The air pressure gage shall be marked to show the normal operating pressure range.

3.7.3 Thermometer. A dial-type, bimetallic thermometer calibrated from 50° F to 500° F shall be furnished. The thermometer shall be installed in a visible location on the kettle and shall be protected by a metal cover. The temperature-sensing element shall be located at the point of greatest heat concentration. The installed thermometer shall be accessible for replacement without the removal of other assemblies and installed attachments not normally removed.

3.7.4 Temperature control. When specified (see 6.2) an automatic temperature control shall be furnished. The temperature control shall be capable of controlling the temperature between plus 100° F and plus 350° F. The temperature sensing element shall be installed at the point of greatest heat concentration. A safety device shall be provided to shut off the burner when the temperature of asphalt cement in the melting tank exceeds the preselected temperature by 15°F.

3.7.5 Lighting torch. A burner lighting torch of the asbestos-wick type with a metal handle not less than 12 inches long and receptacle for storing the torch shall be furnished.

3.8 Hand-spray attachment. The handspray attachment shall consist of a spray tube with spray nozzle(s), handle and control cock, cutoff valve, and a half coupling for connection to the hose. The hose shall be l-inch diameter, 15-feet long, asbestos-lagged, metallic hose with high-temperature fittings. The hose shall be furnished with quick-disconnect couplings on each end. The spray line (pump discharge) on the kettle shall be equipped with a faucet-end half coupling for connection to the hose, and with chain-secured dust cap. The spray nozzle(s) shall have a total capacity of not less than 10 gpm at a pressure of 30 psi. The nozzle(s) shall produce a flat-type spray pattern and shall provide even distribution from a height of not less than 10 inches.

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3.8.1 Carrying and storage rack. Provisions shall be made to store and carry the hand-spray attachment on the kettle when it is not in use. The hand-spray attachment shall remain in its carrying place when the kettle is subjected to towing and shipping requirements specified herein.

3.9 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. A lubrication chart in accordance with SAE J753 shall be furnished with each kettle.

3.9.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, Type I, II, or 111. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a straight hydraulic coupler and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.9.2 Pressure-release device. An automatic pressure-release device shall be provided where pressure-lubricating equipment will damage grease seals or other parts.

3.9.3 Filling, draining, and checking provisions. Enclosures such as gearcases that contain a reservoir or lubricants for the lubrication of the parts enclosed shall be equipped with dipstick, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with lubricant and for draining. The drain outlet shall be fitted with a drainplug conforming to MS35844. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be accomplished when the kettle is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to hand held container.

3.9.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. The kettle shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases may not be compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.9.5 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to the operating level with winter grades of military oil-designated for use in the temperature range of 0° F to 32°F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

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3.10 Fungus and moisture resistance. Unless otherwise specified, (see 6.2) the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I.

a. Components or circuit elements that are inherently fungus and moisture resistant-or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.11 Transportability*

3.11.1 Lifting attachments. Permanently affixed attachments that enable the kettle to be lifted in its normal travel position shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209. Lifting **attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.11.2.**

3.11.2 Tiedown attachments. Permanently affixed attachments that permit fastening the kettle to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.11.1.

3.11.3 Rail humping. When specified (see 6.2) the kettle shall withstand a rail hump test (see 4.5.2*9) without permanent deformation to any component. When commercial facilities are not available to conduct a rail hump test as specified, testing may be conducted at a Government facility.

3.12 Identification marking. The kettle shall be identified in accordance with MIL-STD-130. The markings shall be applied to the kettle on a plate conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.12.1 Shipping data plate. Shipping data plate shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material and in addition, shall show the silhouette of the kettle in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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3.12.2 Control, identification, safety, and warning plates. A plate shall be furnished for each switch, gage, and control and at all points requiring attention for safe and efficient operation of the kettle. The plates shall be labeled by lettering or other appropriate marking with information such as function, direction the control is to move, neutral position, safety instructions, and other warnings or instructions, as necessary. The plates shall be anodized aluminum, sunken-etched or raised-relief. Plates that describe safety measures or plates that warn against operations detrimental to the kettle shall be distinguished by the words "CAUTION" or "WARNING" in red lettering. The plates shall include, but not be limited to the following:

a. Flow of asphalt cement in each position of the 3-way valve.
b. Maximum safe operating pressure of fuel tank.
c. Instructions for releasing fuel-tank pressure at time of kettle shut-down.

The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.13 Electromagnetic interference. Unless otherwise specified (see 6.2) the electromagnetic interference emission and susceptibility characteristics of the kettle shall conform to SAE J551, devices category.

3.14 Treatment and painting. The portions of the kettle normally painted shall be cleaned; treated, and painted in accordance with MIL-T-704. Parts of the kettle subjected to a temperature of 400° F or over shall be painted with heat resistant paint. Unless otherwise specified (see 6.2) the finishing coat shall be lusterless forest green in accordance with MIL-E-52929, Type II.

3.15 Stenciling. The gross weight of each kettle shall be stenciled on each side of the kettle. The prescribed tire pressure shall be stenciled on each fender. "Lift" and "Tiedown", as applicable, shall be stenciled adjacent to their respective tiedown and lifting attachments. All stenciling shall be in blocktype letters not less than 1 inch high.

3.16 Toolbox. The kettle shall be provided with a toolbox made of steel not less than 0.0747 inch in nominal thickness. The toolbox shall have a hinged lid with the hinge extending the full length of one edge and shall be provided with not less than two fasteners that will keep the lid closed when the toolbox is subjected to vibration. The toolbox lid shall have an overlapping lip to prevent entry of rain and dirt. The toolbox shall be securely mounted in a protected and accessible location permitting the lid to be opened through an arc of at least 90 degrees. The lid shall remain in the fully open position by the weight of the lid or by holding devices.

3.17 Workmanship. All parts, components, and assemblies of the kettle, including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All exposed edges shall be rounded or chamfered.

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3.18 Size 1.

3.18.1 Melting tank. The melting tank shall be constructed of steel not less than 0.1345 inch in nominal thickness and shall be jacketed with not less than l-inch-thick insulation conforming to MIL-I-22023, Type I, Class 4. The insulation shall be protected by a steel sheath. A splash plate shall be provided around the inner rim of the melting tank. A draw-off valve that permits emptying the melting tank by gravity shall be provided and shall be located so that the hottest material will be drawn first. The inlet to the valve shall be provided with a removable screen. A dial type, bimetallic thermometer calibrated from 50° to 500° F shall be provided (see 3.7.3). Supports for the heating unit shall be provided within the melting tank. The melting tank shall have a capacity of 80 gallons plus 4 percent for expansion.

3.18.2 Heating unit. The heating unit shall be of the immersion tube type and shall consist of a burner compartment to accommodate the burner specified in 3.7.1, a fire tube, two return flues, and an exhaust stack for each flue. Each exhaust stack shall be provided with a weatherproof cover. Handles, located at each end of the heating unit, shall be provided to facilitate removal and installation of the heating unit. The heating unit shall also serve as a partial cover for the melting tank.

3.18.3 Cover. That portion of the melting tank that is not covered by the heating unit shall be provided with a convex cover mounted on a hinge. The cover shall be made of steel not less than 0.0747 inch in nominal thickness. A device shall be provided that will hold the cover in the fully open position and also at any desired angle between fully open and closed positions. Handles shall be provided to aid in raising and lowering the cover. A latch shall be provided to hold the cover in the closed position.

3.18.4 Pump and hand-spray attachments. The pump shall be mounted on the kettle with the inlet to the pump connected so that it will pump asphalt cement from the bottom of the melting tank.

3.18.5 Engine-driven pump. Unless otherwise specified (see 6.2), the pump shall be driven by a commercial gasoline engine of not less than 3 horsepower. The fuel tank shall be in accordance with 3.19.6.3. The pump shall be of the positive-displacement, gear type and shall discharge not less than 10 gallons per minute (gpm) at 30 psi. The engine and pump shall be completely enclosed in a metal housing in accordance with 3.19.6.6

3.18.6 Manually-operated pump. When specified (see 6.2), the pump shall be manually operated and shall be of the positive-displacement, gear type. The pump shall discharge not less than 10 gpm when operated at normal cranking speed.

3.19 Size 2.

3.19.1 Melting tank. The melting tank shall be constructed of sheet steel not less than 0.1345 inch in nominal thickness. Seams shall be welded and

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leaktight. Splash plates shall be provided on the top edges. The melting tank shall be bolted to the furnace shell so as to be removable. Two removable, perforated-steel plates shall be provided in the melting tank for separating the cold material from the melted material. When in position, the plates shall form an inverted V and shall run the full length of the tank. An outlet shall be provided at the lowest point in the tank and shall be fitted with a l-1/2-inch, 3-way valve arranged for connecting the kettle to the pump, to the hand-spray attachment, or to a drain valve for draining the melting tank by gravity. The melting tank shall have a capacity of 165 gallons plus 4 percent for expansion.

3.19.2 Furnace shell. The furnace shell shall be constructed of sheet steel not less than 0.1046 inch in nominal thickness. All seams shall be welded. The top edges shall be flanged to form a support for the melting tank. Baffle plates and a vent stack made of steel shall be provided. The baffle plates shall be bolted in place and shall be arranged to provide heat circulation completely around the sides and the bottom of the melting tank. There shall be no vent holes in the sides of the furnace shell. The vent stack shall be bolted in place and shall be provided with a hinged cover. The furnace shell shall have a burner compartment to accommodate the burner specified in 3.7.1.

3.19.3 Cover. A hinged cover made of steel not less than 0.0747 inch in nominal thickness shall be provided on the melting tank. A device shall be provided that will hold the cover in the fully open position and also at any desired angle between fully open and closed position. The cover shall have a hinge extending the full length of one edge and shall be provided with not less than two fasteners that will keep the cover closed. The cover shall have an overlapping lid to prevent entry of rain and dirt. Handles shall be provided to aid in raising and lowering the cover. When specified (see 6.2), a fusible metal link (heat-sensitive) for holding the cover open shall be provided and shall be incorporated in the cover holding mechanism to permit gravity closing of the lid in event of fire.

3.19.4 Grille. A removable grille, which shall support a 55-gallon drum while being emptied into the melting tank, shall be provided under the cover.

3.19.5 Bitumen pump and control valve.

3.19.5.1 Pump. The kettle shall be provided with a removable, gasoline-engine-driven bitumen pump. The pump shall deliver asphalt cement having a penetration value between 85 and 100, as determined by the method specified in ASTM D-5, through the spray nozzles at" a rate of not less than 10 gpm at not less than 30 psi. A bypass relief valve shall be provided for the pump. The pump shall be installed on the kettle and connected to permit the following operations:

a. Fill the kettle from a supply tank.
b. Circulate the asphalt cement being heated in the kettle.
c. Pump the asphalt cement through the hand-spray attachment.
d. Pump cleaning solvent through the hand-spray attachment without the solvent entering the melting tank.

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The valves used to accomplish the above operations shall be ball or plug type valves. The material flow through the valve ports shall be indicated by distinct markings on the valve control stem. An instruction plate (see 3.12.2) shall be located near the valve control and the plate shall show the proper valve setting to perform the above specified operation. The plates shall be attached as specified herein.

3.19.6 Gasoline engine.

3.19.6.1 Commercial engine. The commercial engine shall be a 2- or 4-cycle, air cooled, gasoline engine equipped with components and accessories necessary for the required performance of the kettle. The engine shall be of current standard production which has been proven commercially acceptable for like applications and for the type of service specified herein. The engine shall be provided with an automatic-rewind rope starter. The rope shall be galvanized flexible steel stranded wire rope or nylon.

3.19.6.2 Power and speed rating. The power and speed rating of the engine shall be such that it shall provide sufficient power and torque to operate the kettle under any of the operational requirements and conditions specified herein, without exceeding the engine manufacturer's published continuous horsepower rating, or 85 percent of the manufacturer's published maximum horsepower rating, at the applicable speed.

3.19.6.3 Engine fuel tank. A fuel tank conforming to the contractor's commercial standard except as specified herein shall be provided. The tank capacity shall be sufficient for 4 hours of operation at the maximum fuel rating of the engine. The tank and all lines shall be located or positioned to eliminate fire hazard caused by spilling, overflow, or drainage of fuel. The tank shall be stenciled: "FILL WITH GASOLINE ONLY". When a two cycle engine is used, the tank shall be marked indicating the gasoline and oil mixture.

3.19.6.4 Gear reduction and clutch. The pump shall be driven through a base-mounted gear reduction and a clutch. The clutch shall be a manually-operated plate or disk type and shall be mounted between the gear reduction and the pump. The clutch shall transmit not less than 150 percent of the maximum torque developed by the engine. All components of the gear reduction and clutch shall be accessible for lubrication, adjustment, and installation of repair parts.

3.19.6.5 Quick-detachable couplings. Quick-detachable couplings shall be provided on the pump discharge, drain valve, hose ends, and handspray attachment. The couplings shall be of the ground joint type.

3.19.6.6 Housing. The engine, gear reduction, clutch and pump shall be completely enclosed in a metal housing. The housing shall have an access lid that permits access to the engine, gear reduction, clutch and the pump. The lid shall be hinged with the hinge extending the full length of one edge and shall be provided with not less than two fasteners that will keep the lid closed when the housing is subjected to vibration. The lid shall be held in the open position by the weight of the lid or by holding devices.

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4* QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapprove-d by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Preproduction inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 Preproduction inspection.

4.3.1 Examination. The kettle shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction kettle shall be tested as specified in 4.5.2 through 4.5.2.12 for size 1 or size 2, as applicable. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each kettle shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each kettle shall be tested as specified in 4.5.2.13. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The kettle shall be examined for the following defects:

101. Dimensions not as specified.
102. Assembly not complete or not correct.
103. Parts or components missing or do not function as specified.
104. Safety devices, guards, insulation, not as specified.

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105. Material not as specified or not in compliance with MIL-STD-888.
106. Ease of maintenance not as specified.
107. Pump not as specified.
108. Engine and engine accessories not as specified.
109. Clutch and gear reduction not as specified.
110. Lights and reflectors not as specified or incorrectly mounted.
111. Heating system not as specified.
112. Hand-spray attachment not as specified.
113. Lubrication devices not accessible, missing, or not installed as specified.
114. Fungus and moisture resistance not as specified.
115. Tiedown and lifting attachments not as specified.
116. Identification markings missing or not as specified.
117. Treatment and painting not as specified.
118. Transportation-data plates and operating instruction plate not as specified.
119. Stenciling not as specified.
120. Repair parts and tools missing or not as specified.
121. Toolbox missing or not as specified.
122. Chassis and components not as specified.
123. Workmanship not as specified.
124. Finish plant not adequate.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The kettle shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.5.2.2 Tank capacity. Determine the capacity of the melting tank. A melting tank capacity less than that specified for the applicable size shall constitute failure of this test.

4*5.2.3 Grille (size 2). Place a 55-gallon drum, filled with liquid asphalt, on the grille and drain the asphalt into the melting tank. Bends, breaks, or permanent deformation of the grille or any part of the kettle shall constitute failure of this test.

4.5.2.4 Burner. Ignite and operate the burner for 1 hour. Determine the amount of fuel consumed by the burner. Consumption of less than 2.5 gallons of fuel, malfunction of the burner or burner controls, or smoking shall constitute failure of this test. To prevent damage to the furnace shell remove the burner before conducting this test.

4.5.2.5 Performance (size 1). Fill the kettle with asphalt cement (chunk type). Ignite the burner and continue to add asphalt cement until the melting tank is filled to capacity. Continue heating until the asphalt cement reaches

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pumping temperature. Determine the capacity of the pump while pumping the asphalt cement. A pump capacity of less than specified in 3.18.5 or 3.18.6 shall constitute failure of this test.

4.5.2.6 Performance (size 2). Fill the melting tank with asphalt cement having a penetration value between 85 and 100 as determined by the method specified in ASTM D5. Allow the asphalt cement to cool for a period of not less than 12 hours. Ignite the burner(s) and heat the asphalt cement until it reaches pumping temperature During warm-up, set controls to circulate the asphalt cement being heated in the melting tank. When the asphalt cement reaches spraying temperature, set controls to pump asphalt cement from the melting tank through the hand-spray attachment and return it to the melting tank. Operate the kettle to pump asphalt cement through the hand-spray attachment for a total period of not less than 8 hours. The asphalt cement shall be maintained at the recommended spraying temperature range. upon completion of the 8 hours, allow the asphalt to cool for 24 hours. Operate the kettle to demonstrate the ability to accomplish the requirements listed below without repair or replacement of any parts. Inability to accomplish any requirement specified herein shall constitute failure of this test. Any evidence of leakage of asphalt cement, fuel, or gear oil or permanent deformation of any part of the kettle shall constitute failure of this test.

a. Set the automatic temperature control (if applicable) and control the temperature of the asphalt cement in the melting tank within the limits specified herein. (Test operation of control when set 250°F, 300°F and 350°F.)
b. Circulate the asphalt cement being heated in the melting tank without the hand-spray attachment.
c. Pump asphalt cement through the spray nozzles at a rate of 10 gpm.
d. Produce 30 psi at the spray nozzles.
e. Pump asphalt cement into the melting tank from an outside source.
f. Pump cleaning solvent through the hand-spray attachment without the solvent entering the melting tank.

4.5.2.7 Test procedures of lifting provisions.

a. Lift the kettle and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the kettle. Determine the forward angles of application for each provision. Measure the clearance between each sling and the kettle. Measure eye openings and clearance dimensions of the provisions. Measure the load carried by each provision.
b. Restrain the kettle by anchoring the main frame and subject each provision to a force equal to the load factor specified in MIL-STD-209 times the load it carried in paragraph (a) above. An alternate method of test is to lift the kettle as in (a) above and add weights to the main frame until the required force is obtained. Hold the kettle under either of the above loading conditions for at least 90 seconds.

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4.5.2.7.1 Failure criteria. Weld failure or permanent deformation caused by lifting or failure to meet the requirements of MIL-STD-209, shall constitute failure of this test.

4.5.2.8 Test procedures of tiedown provisions. With the kettle anchored by the tiedown attachments, apply static loads as specified (see 3.11.2) to each attachment. Hold each applied force for a minimum of 90 seconds.

4.5.2.8.1 Failure criteria. Weld failure or permanent deformation of the kettle or its tiedown attachments or damage to any tiedown cable shall constitute failure of this test.

4.5.2.9 Railroad humping. When the rail hump test is specified (see 6.2) anchor the empty kettle on a domestic service standard railway flat car. Blocking shall be as specified by the kettle manufacturer. Hump the car, with the kettle tied down, at a speed of not less than 8 mph against two standing rail road cars with their brakes set. The kettle shall face in the direction of the two cars. Repeat the procedure with the rear of the kettle in the direction of the two cars. The kettle may be reversed by either reloading it in the proper direction on the flat car or by reversing the direction of the flat car. Once the test has begun, there will be no readjustment of load, or any reconditioning of the blocks, chocking material or tiedowns except to reverse direction of the kettle. After each impact, the kettle and tiedown shall be inspected to determine permanent deformation to components of the kettle, or other evidence of possible failure. After completion of the railroad impact test, an operational checkout of the kettle will be conducted. The operational checkout shall include heating, circulating and spraying asphalt cement.

4.5.2.9.1 Failure criteria. Weld failure, or permanent deformation to the components of the kettle or its tiedown attachments or damage to any tiedown cable or stopping of a test due to impending failure and possible damage to the kettle, or failure of the kettle to operate after the test shall constitute failure of the test.

4.5.2.10 Road. Fill the melting tank to capacity with asphalt cement and fill the burner fuel tank to capacity with fuel. Couple the kettle to the towing vehicle specified herein and perform the road test specified in table IV.

TABLE IV. Road test.

Course	Mileage and Speed
Primary highway	100 miles at variable speeds up to 45 mph.
Secondary road	100 miles at variable speeds up to 25 mph.

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During the road test perform and record the actual results of the following:

a. Tow the kettle over 5-inch high obstacles so that all combinations of the wheels have passed over the obstacles. Observe tire-to-surface contact, evidence of binding or interference, and after returning the suspension to normal, measure for evidence of misalinement.
b. Measure and record the deviation or off tracking of the trailer with respect to the prime mover, when the kettle is towed straight ahead on a paved surface.
c* Jackknife the kettle at angles up to 60 degrees from straight ahead both to the left and to the right.
d. Observe the kettle electrical system components operation during the road test.
e. Inspect the hubs and bearings of at least one wheel for damage after completion of all required road tests.
f. Operate after completion of test.

4.5.2.10.1 Failure criteria. Failure to complete the test, inability to operate and function as specified herein upon completion of the test, accessories damaged or not remaining in their carrying places, permanent deformation or damage to the kettle or trailer, or evidence of instability while being towed shall constitute failure of this test.

4.5.2.11 Electromagnetic interference. The preproduction model shall be tested to determine conformance with 3.13. The contractor shall furnish the contracting officer a report of tests required to prove conformance to SAE J551. Disapproval of the report shall constitute failure of this test.

4.5.2.12 Sound level measurement. Measure the sound level at the operators station in accordance with SAE J919.

4.5.2.12.1 Failure criteria. Failure to meet the criteria specified in 3.4.4 shall constitute failure of this test.

4.5.2.13 Quality conformance testing. Operate each kettle using not less than SAE 50 engine oil for run-in period of not less than 30 minutes. Check the operation of all controls, and make necessary adjustment for proper operation. Any leak, inability of any control to operate as required, or malfunction of any component that is not corrected by adjustment shall constitute failure of this test.

4.6 Inspection comparison. The Government may select kettles at any time during the contract production period and subject these kettles to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and

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will not include the previously inspected preproduction model kettles. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison kettle to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison kettle(s) and shall be cause for refusal by the Government to continue acceptance of production kettles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on kettles previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all kettles accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation, packing, and marking shall be examined in compliance with the Quality Assurance Provisions of MIL-D-771.

5* PACKAGING

5.1 Preservation, packing and marking. Each complete kettle with accessories, repair parts, tools and publications shall be preserved, packed and marked in accordance with MIL-D-771. The level of preservation shall be level A, B, or commercial and the level of packing shall be level A, B, or commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The kettle is intended for use in transporting, heating and spraying all types of bituminous material and tar for maintenance of roads and runways.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. When a preproduction model is not required (see 3.2 and 4.3).
e. Time frame for submission of production model and number of kettles required (see 3.2).
f. When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

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g. When a fire extinguisher will be furnished by the Government (see 3.4.2).
h. When a vehicular electrical system is required (see 3.6.9 and 3.6.14).
i. When an intervehicular connector is required (see 3.6.9).
j. When a 24-volt vehicular electrical system is required (see 3.6.14).
k. When an automatic temperature control is required (see 3.7.4).
l. When fungus and moisture resistance is not required (see 3.10).
m. When railroad hump test is required (see 3.11.3 and 4.5.2.9).
n. When electromagnetic interference will be other than as specified (see 3.13).
o* When color required will be other than lusterless forest green (see 3.14).
p" When manually operated pump is required (see 3.18.5 and 3.18.6).
q. When a fusible metal link is required (see 3.19.3).
r. Level of preservation and level of packing required (see 5.1).

6.3 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the kettle covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.4 Government-furnished property. When a fire extinguisher is required the contracting officer should arrange to furnish a 2-1/2 pound dry chemical fire extinguisher with bracket (see 3.4.2) for each kettle being procured.

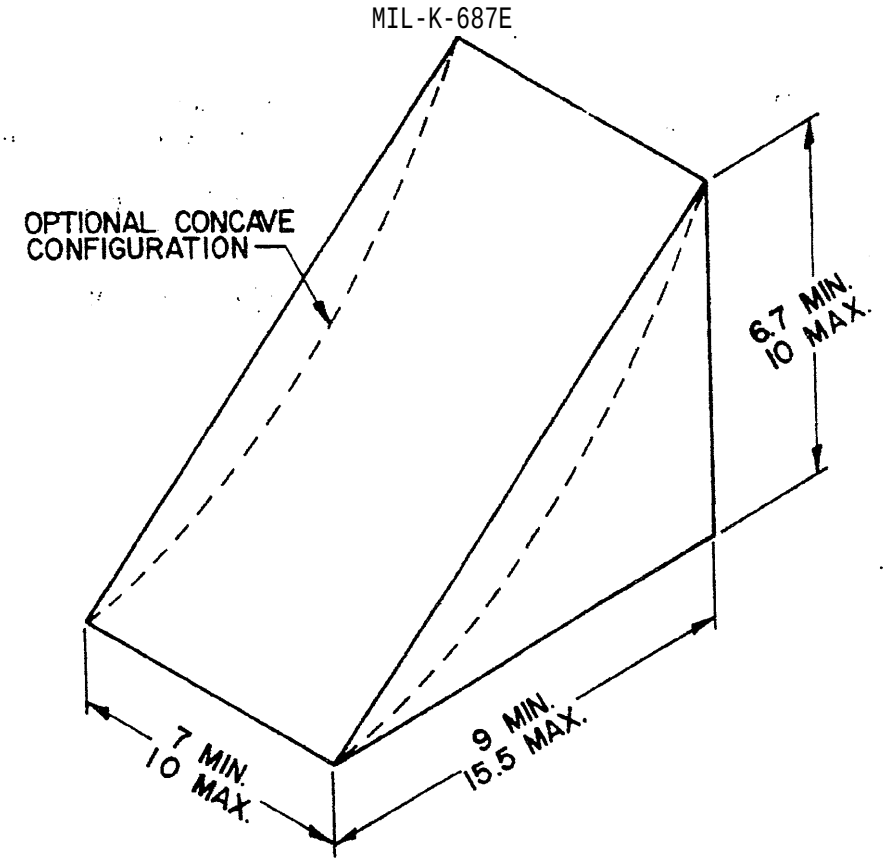
6.5 Preproduction model. Any changes or deviations of production kettles from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish kettles conforming to this specification.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list, and contractor's maintenance and operation manual to be furnished with each kettle.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of kettles. A suggested paragraph is as follows:

"Shipment of kettles shall include repair parts, maintenance tools, operational instructions, and accessories; unless exceptions are provided elsewhere in the contract."

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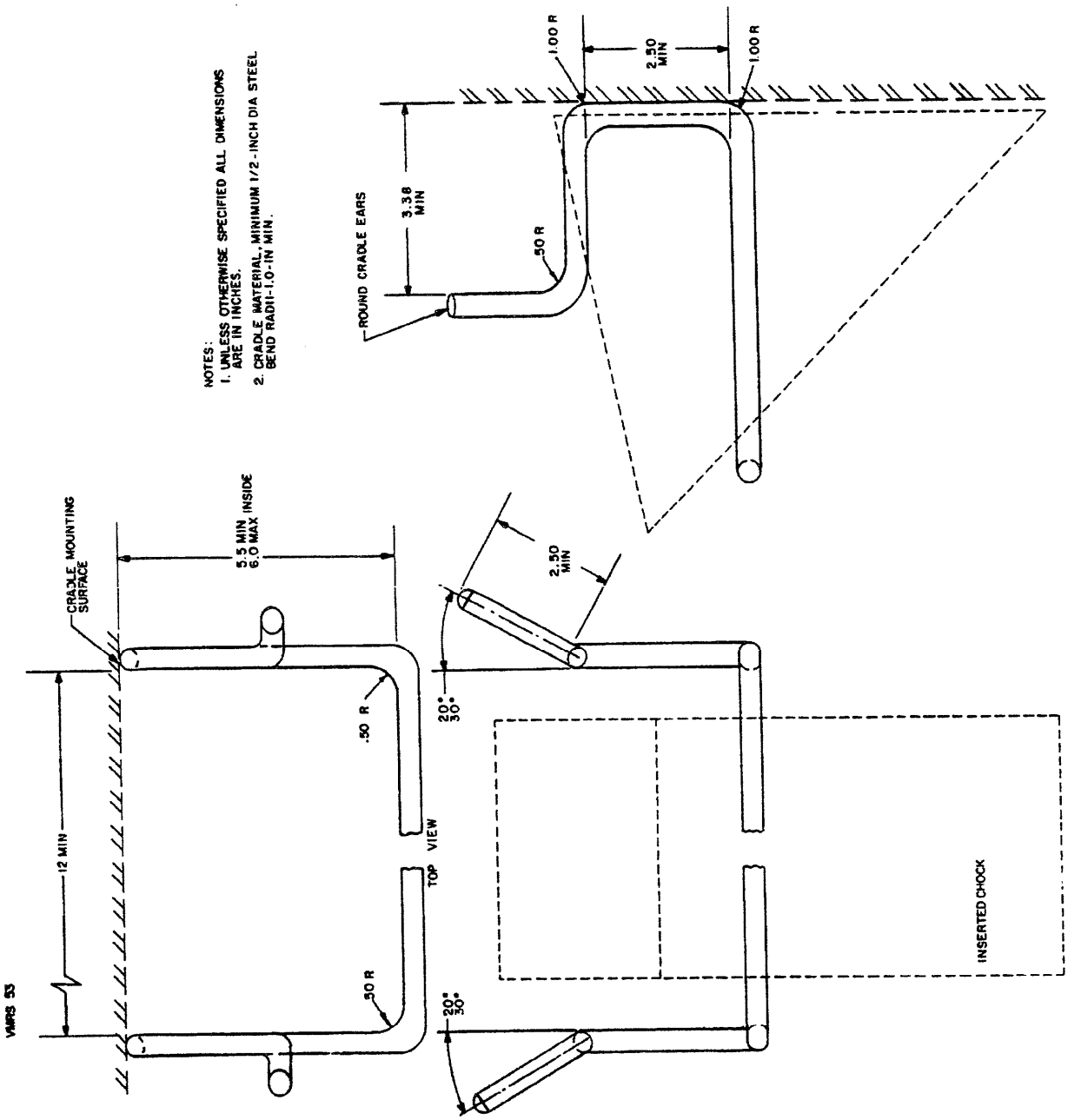
1. ALUMINIUM
2. MAGNESIUM
3. SEMI-RIGID ELASTOMER
4. MAXIMUM CHOCK WEKMT-11.5 LB.

CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 1. Chock block assembly,

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X-604B

FIGURE 2. Cradle.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-K-687E Kettles, Heating, Bitumen, Trailer-Mounted

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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